Superfluid + AI Agent Skills

Teaching AI to speak Superfluid

Kaspar Kallas
Software Product Engineer
10+ years in software, 4 with Superfluid

GoodBuilders Webinar

Agenda

01 Superfluid Protocol Overview

02 Superfluid Clear Signing

03 Brief History of AI-Driven Engineering

04 Agent Skills: What & Why

05 The Superfluid Skill: Deep Dive

01

Superfluid Protocol Overview

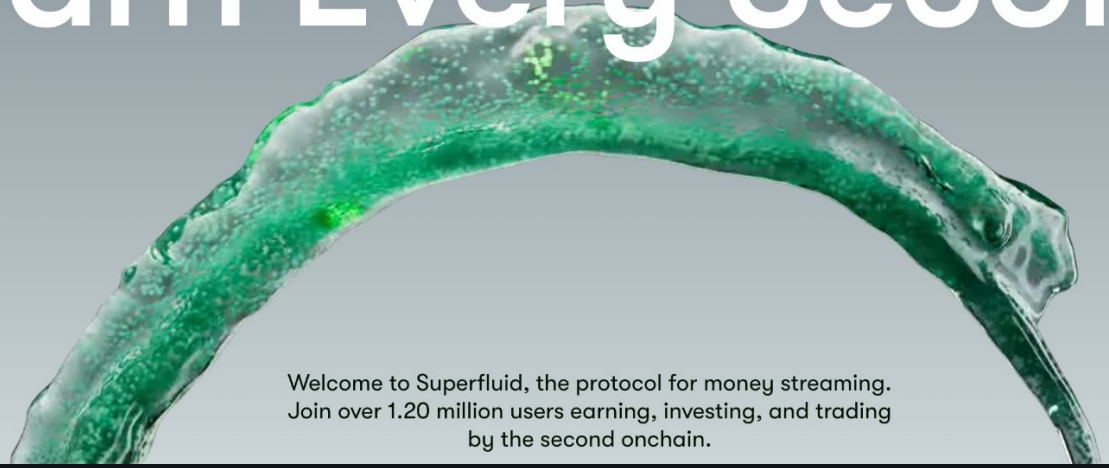
Real-time finance on the blockchain

Superfluid

Technology Builders Adoptions Ecosystem Governance Contact

EXPERIENCE THE FUTURE OF MONEY

Earn Every Second



Welcome to Superfluid, the protocol for money streaming.
Join over 1.20 million users earning, investing, and trading
by the second onchain.

superfluid.org

What is Superfluid?

Money Streaming

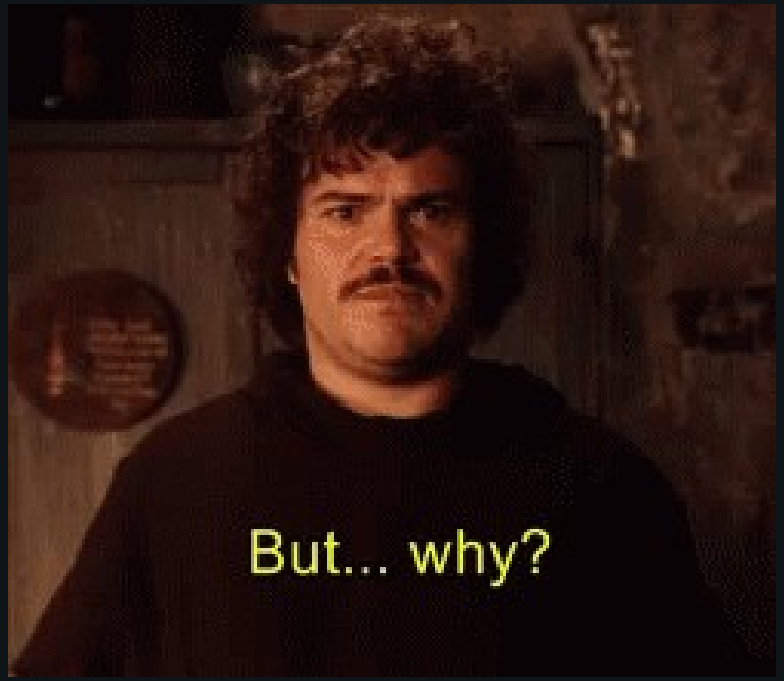
Instead of discrete transfers, Superfluid enables continuous, per-second token flows.

Think of it as a pipe that moves tokens every single second. No repeated transactions needed.

Key Properties

- Per-second settlement
- No recurring transactions
- One tx to start, one to stop
- Programmable via SuperApps
- Multi-chain deployment

Money that moves every second. No transactions. Just... flows.



What can you actually do with money streams?

Use Cases for Money Streaming

Recurring Payments

Salaries, payroll, subscriptions, recurring donations

Token Distribution

Staking rewards, airdrops, vesting, UBI (GoodDollar)

Financial Splitting

Revenue sharing, royalty splitting, DAO treasury diversification

DeFi

Streaming DEXes, dollar-cost averaging, yield streaming

Access Control

Pay-per-second API usage, stream-gated subscriptions

Creator Economy

Patronage streams, royalty pools for NFT holders

DAO & Governance

Contributor compensation, grant streaming

Gaming

Play-to-earn streams, in-game resource streaming, asset rental

Super Tokens

ERC-20 superset with programmable financial primitives

Wrapper Super Tokens

- Wraps an existing ERC-20
- Upgrade / downgrade between the original token and the Super Token
- e.g. USDCx, DAIx, ETHx

Pure Super Tokens

- Native Super Token, no underlying ERC-20
- Born streamable
- **G$ is a Pure Super Token**

Modular Token Infrastructure

The Superfluid Protocol is extended through composable modules called Agreements

Agreements as Modules

- Each Agreement adds a new financial primitive to the token
- Adding modules expands what the token can do
- The protocol is designed so new Agreements can be added over time

Agreements

CFA

Constant Flow Agreement

- 1-to-1 continuous streams
- Sender to Receiver at X tokens/second
- Create, update, or delete flows

GDA

General Distribution Agreement

- Many-to-many distributions
- Pool-based: distribute to all members proportionally
- Instant or streaming distributions

Batch Calls

Execute multiple Superfluid operations in a single transaction

Super Token-based batching

- Combine multiple Super Token operations into one transaction
- Create flows, distribute tokens, transfer, unwrap, all atomically
- Simplifies complex multi-step workflows

Superfluid pioneered token-based batch calls early on. The pattern has since been popularized by account-based batch calls in smart wallets like Gnosis Safe.

SuperApps

Smart contracts that react to streaming events

Programmable Hooks (Callbacks)

`onCreate`

Triggered when a new flow is created to the SuperApp

`onUpdate`

Triggered when an existing flow is updated

`onDelete`

Triggered when a flow is deleted

Composable building blocks: chain SuperApps together to build complex logic on top of streams

02

Superfluid Clear Signing

Making transactions human-readable

Blind Signing vs Clear Signing

Blind Signing

- User sees raw hex data
- No idea what they're approving
- Trust the dApp completely
- Phishing & exploitation risk

Clear Signing (EIP-712)

- Typed, structured data
- Human-readable transaction content
- Verify exactly what you're signing
- **Better UX for everyone, not just AI**

Superfluid Clear Signing

Ready and functional, pending mainnet deployment

What it enables:

- Users can read and verify Superfluid operations before signing
- "Create flow of 100 DAIx/month to 0xABC..." instead of raw calldata
- Great UX improvement for all users, regardless of AI involvement
- Even more important when an AI agent initiates a transaction on your behalf

Segue: so how do we get AI to understand Superfluid correctly in the first place?

Traditional Signing

Confirm Transaction

To Contract

`0xd399e2Fb5f4cf3722a11F65b88FAB6B2B8621005`

Function

`batchCall((uint256,address,bytes)[])`

Operations

2 ERC20_TRANSFER_FROM

201 CALL_AGREEMENT (createFlow)

Calldata 804 bytes

```
0x862bb56d0000000000000000000000000000000000000000000000000000000000000002
0000000000000000000000000000000000000000000000000000000000000200000000
0000000000000000000000000000000000000000000000000000000000000400000000000000000
0000000000000000000000000000000000000000000000000120000000000000000000000000
0000000000000000000000000000000000000000000000000200000000000000000000000131780640ed
```

Gas Estimate ~0.003 ETH

Connect Wallet

Click to see what your wallet shows for a raw batch call

Clear Signing

English

Sign Transaction Send & Stream

You are about to send fUSDCx and start a continuous money stream

EIP-712 Message

Send 10 fUSDCx and stream 10 fUSDCx/month to 0x42bb...afd7

To	0x42bb ... Afd7
Token	fUSDCx
Send Amount	**10 fUSDCx**
Flow Rate	**9.9999 fUSDCx per month**

This will send fUSDCx and continuously stream tokens until you cancel

Connect Wallet

Users see exactly what will happen, in their own language

clear-signing-website.vercel.app/demo

03

Brief History of AI-Driven Engineering

The Evolution

Search & Autocomplete

StackOverflow, docs, language-aware autocomplete in IDEs

StackOverflow, IDE autocomplete

>

Code Completions

AI predicts and suggests the next line or block of code

GitHub Copilot, Cursor Tab

>

Coding Agents

AI plans, implements, and reviews in rapid loops

Claude Code, OpenCode, Codex

>

Autonomous Agents

AI operates tools, makes decisions, executes end-to-end

OpenClaw, Ralph loop

The Context Problem

LLMs are general-purpose but shallow on specific protocols

- They hallucinate ABIs and function signatures
- They use outdated patterns and outdated tools
- They're unaware of new tools and recent SDK changes

Always test and verify AI-generated code, especially in Web3 where mistakes can mean lost funds.

04

Agent Skills
What & Why

What are Agent Skills?

Markdown-based knowledge packages (SKILL.md) that teach AI agents domain expertise on demand. Can also include simple scripts and helper tools.

Progressive Disclosure

Metadata scan

~100 tokens

>

Full instructions

<5k tokens

>

Reference files & scripts

On demand

The Ecosystem

Supported Agents

Claude Code, Codex, Gemini CLI, Cursor, GitHub Copilot, Cline, Windsurf, OpenCode, and more

Skills = Knowledge

Teach AI what it needs to know: protocols, patterns, ABIs, docs, and helper scripts

MCP = Tool Execution

Give AI access to APIs via API keys, databases, on-chain actions, external services

05

The Superfluid Skill Deep Dive

github.com/superfluid-org/skills

Why Build a Superfluid Skill?

LLMs know about Superfluid... kind of.

- Training data is outdated or incomplete
- Superfluid has gone through 4 SDK generations
- AI uses outdated patterns and isn't aware of new tools
- Contract interfaces are complex and protocol-specific

The skill bridges this gap: verified, current, structured knowledge that loads when needed.

AI + Smart Contracts

Why the ABI alone is not enough

The ABI gives you a structured interface

- Every contract comes with an ABI, so AI works decently
- But the ABI lacks nuance and detail
- Public function does not mean callable in any context
- Functions have access patterns, gotchas, and weird behaviours
- Immutable deployments: mistakes are hard to remedy

Rich ABI: the Superfluid Skill approach

- Context and detail on top of ABI signatures
- Access patterns: who can call what, when
- Events and errors each function emits
- Common gotchas and edge cases
- Human-readable descriptions

ABI vs Rich ABI

Same function, different level of understanding

Standard ABI (JSON)

```
{
  type: 'function',
  name: 'createFlow',
  inputs: [
    { name: 'token',
      type: 'address' },
    { name: 'receiver',
      type: 'address' },
    { name: 'flowRate',
      type: 'int96' },
    { name: 'ctx',
      type: 'bytes' },
  ],
  outputs: [{ name: 'newCtx',
    type: 'bytes' }],
  stateMutability: 'nonpayable'
}
```

Rich ABI (YAML)

```
# == Flow Management ==
# Functions with ctx are called
# through the Host...
createFlow:
  # Start a new stream from
  # ctx.msgSender to receiver.
  mutability: nonpayable
  access: anyone  # ctx.msgSender
  inputs:
    - token: address
    - receiver: address
    - flowRate: int96  # > 0
    - ctx: bytes
  emits:
    [FlowUpdated, ...]
  errors:
    [CFA_ZERO_ADDRESS_RECEIVER,
     CFA_INVALID_FLOW_RATE, ...]
  notes:
    - "Deposit taken as safety
       margin for solvency."
```

What's Inside

v0.6.0, growing continuously

Protocol Concepts

Architecture and core mechanics distilled for AI

Rich ABI References

ABIs enriched with access patterns, gotchas, context

Subgraph & Indexing

Query patterns and schema references

SDK Guidance

Navigation across Superfluid's SDK evolution

SuperApp Patterns

Hooks, callbacks, and composability

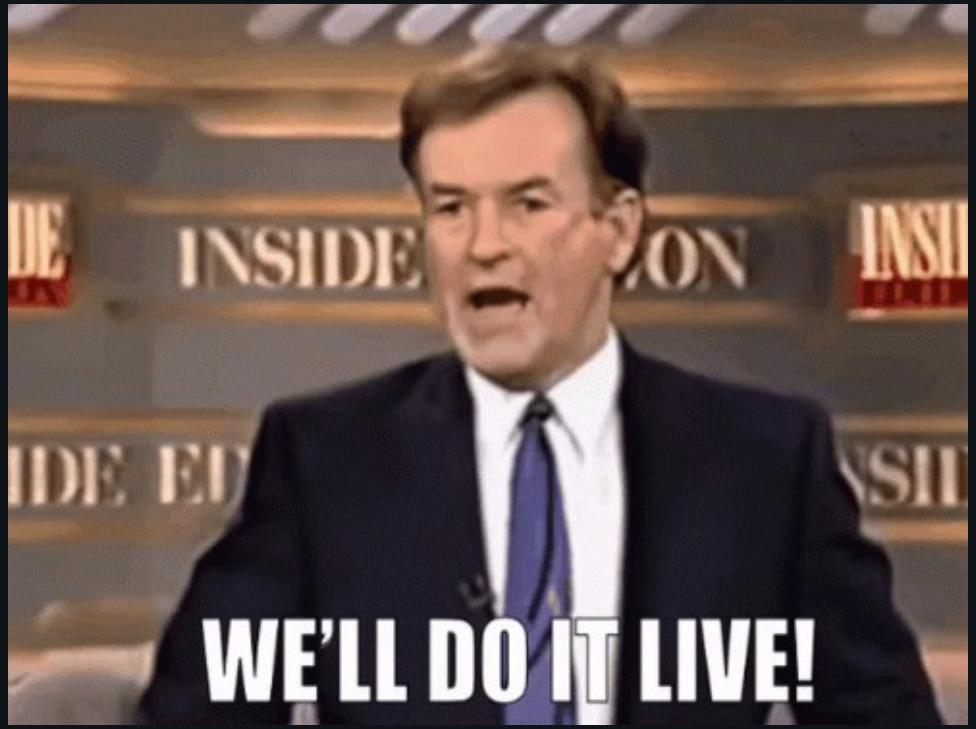
Helper Scripts

Address resolution, token lists, balances

Pre-Recorded Demo

Without the skill > With the skill

[Replace this slide with your screen recording]



Live Demo

Get Started

```
$ npx skills add superfluid-org/skills
```

Works with Claude Code, Codex, Cursor, Copilot, Gemini CLI, and 10+ more agents

Links

skills.superfluid.org
github.com/superfluid-org/skills
skills.sh/superfluid-org/skills/superfluid

Alternative Installs

Claude Code plugin marketplace
ClawHub (for OpenClaw)
ZIP upload for Claude Desktop

Superfluid Skill

for AI Agents

Real-time finance knowledge for your AI agent

A complete Superfluid Protocol reference: contracts, SDKs, subgraphs, and more.

v0.6.1

GitHub skills.sh ClawHub

bash — superfluid-skill

```
# Install for Claude Code, Codex, OpenCode, OpenClaw, Cursor, GitHub
Copilot, Cline, Windsurf, Gemini, and 10+ more agents
$ npx skills add superfluid-org/skills

# Update the skill later
$ npx skills update

# — ALTERNATIVE INSTALL METHODS —
# If you prefer not to use skills.sh

# Run inside OpenClaw
$ npx clawhub@latest install superfluid

# Run inside Claude Code
$ /plugin marketplace add superfluid-org/skills
$ /plugin install superfluid@superfluid-skills
```

skills.superfluid.org

What Can You Do With It?

With AI that actually understands Superfluid, you can:

- Brainstorm your next app idea
- Ask and debug difficult protocol-specific problems
- Automate common observational tasks with natural language
- Quickly create demos, UIs, and prototypes that integrate Superfluid

Install it. Try it. Build something.

Bonus: Built Overnight by an Autonomous Agent

A fully autonomous AI agent re-implemented Superfluid Explorer using the Superfluid Skill

SF Explorer | Celo | Search... ⌘K

Celo (42220) | Home | Tokens | Protocol | Liquidations | Super Apps

Celo

Network overview and latest activity

Active Streams	Active Pools	Listed Super Tokens
6,165	13	7
Total number of active streams across all tokens	Total number of active distribution pools	Number of Super Tokens listed on the network

Latest Streams

Sender	Receiver	Token	Flow Rate	Total Streamed	Created
djlethalskillz.eth 0x5aaE...cECF	0xf3d6...7312	G$	49.31506 G$/mo	3.67437	2d ago
djlethalskillz.eth 0x5aaE...cECF	0xE4f6...E7Bc	G$	49.31506 G$/mo	3.6749	2d ago
xzeroz97 0xf9ff...b63F	0xC1dC...Ea9c	G$	0 G$/mo	55.03323	2d ago
0x2B46...754F	0x1866...52a2	G$	4,931.50684 G$/mo	540.02853	3d ago
0x2B46...754F	0x1866...52a2	G$	0 G$/mo	10.70776	3d ago
0xBC61...7264	0x43d7...a4A1	G$	49.99999 G$/mo	6.04627	3d ago

superfluid-explorer-v2-by-ai-agent.vercel.app

Superfluid + AI Agent Skills

github.com/kasparkallas/superfluid-explorer-v2-by-ai-agent

Q & A

skills.superfluid.org

github.com/superfluid-org/skills